

PURCHASE DESCRIPTION

WASHCLOTH, MICROFIBER

This purchase description is approved for use by all Departments and Agencies of the Department of Defense. (DoD).

1. SCOPE

1.1 Scope. This purchase description covers the requirements for a terry microfiber washcloth treated with a soil release finish and is intended for use by military personnel of the Department of the Army.

1.2 Classification. The washcloth will be available in one type and size.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-4 - Glossary of Fabric Imperfections

Comments, suggestions, or questions on this document should be addressed to: Department of the Army, Natick Soldier Research, Development and Engineering Center, Kansas St., Natick MA 01760. ATTN: RDNS-WPW-C.

FSC 7210

DISTRIBUTION STATEMENT A. approved for public release; distribution is unlimited.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Test Method 8 - Colorfastness to Crocking: AATCC Crockmeter Method
AATCC Test Method 15 - Colorfastness to Perspiration
AATCC Test Method 16 - Colorfastness to Light
AATCC Test Method 61 - Colorfastness to: Laundering: Accelerated
AATCC Test Method 70 - Water Repellency: Tumble Jar Dynamic Absorption Test
AATCC Test Method 81 - pH of the Water-Extract from Bleached Textiles
AATCC Test Method 130 - Soil Release; Oily Stain Release Method
AATCC Test Method 135 - Dimensional Change of Fabrics after Home Laundering
AATCC Evaluation Procedure 1, Gray Scale for Color Change
AATCC Evaluation Procedure 2, Gray Scale for Staining
AATCC Evaluation Procedure 8, AATCC 9-Step Chromatic Transference Scale
AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles, Option A
AATCC Stain Release Replicas

(Copies are available for purchase on-line at <http://www.aatcc.org/> or from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies are available on-line at <http://www.asq.org> or from the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

ASTM INTERNATIONAL

ASTM D 629 - Standard Test Method for Quantitative Analysis of Textiles
ASTM D 3775 - Standard Test Method for Fabric Count of Woven Fabric
ASTM D 3776 - Standard Test Method for Mass Per Unit Area (Weight) of Fabric, Option C
ASTM D 3787 - Standard Test Method for Bursting Strength of Textiles-Constant-Rate-of-
Traverse (CRT) Ball Burst Test

(Copies are available for purchase on-line at <http://www.astm.org/>. or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19426-2959.)

OTHER PUBLICATIONS

Repeat Insult Patch Test - Modified Draize Procedure - Principles and Methods of Toxicology, (4th edition) A Wallace Hayes (editor), pp 1057-1060, 2001

(Copies are available on-line at <http://www.taylorandfrancis.co.uk> or from Taylor and Francis, 325 Chestnut Street, Philadelphia, PA 19106.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract in the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Recycled, recovered or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design. The finished dimensions of the washcloth shall be 12-1/4 inches by 12-1/4 inches ($\pm 1/8$) inch. The side edges (border) shall be firm selvages and the top and bottom rounded edges shall be serged and shall finish 3/16 ($\pm 1/16$) inch wide. As an alternate all edges may be serged.

3.4 Materials.

3.4.1 Basic material. The basic material for the washcloth shall be a 100 percent polyester terry microfiber (400 filament – 48 filaments warp per inch) warp knitted with polyester scrim (see 6.5). The finished cloth shall not present a dermal health hazard when used as intended (see 3.12) and shall conform to the physical requirements of Table I when tested as specified below.

TABLE I. Physical requirements

Characteristic	Requirement/ Requirement Paragraph	Test Method
Fiber Content	3.3.1	ASTM D 629
Weight, oz/sq yd (per dozen)	12 $\pm$.50	ASTM D 3776
Yarns per Inch		
Warp	12	ASTM D 3775
Weft	15	ASTM D 3775
Bursting Strength	90	ASTM D 3787
Visual Shade Matching	para. 3.5.1	AATCC Evaluation Procedure 9, Option A <u>1</u> /
Colorfastness:	para. 3.5.2	AATCC 61, Opt. 2A
Laundering (after 3 cycles)	para. 3.5.2	AATCC 16, Opt 1 or 3
Light (after 40 hrs or 170 kJ)	para. 3.5.2	AATCC 15
Perspiration (acid and alkaline)	para. 3.5.2	AATCC 8
Crocking		

TABLE I. Physical requirements (cont'd)

Characteristic	Requirement/ Requirement Paragraph	Test Method
Dimensional Stability (3 cycles)	para. 3.6	AATCC 135 <u>2/</u>
Wicking	para. 3.7	<u>3/</u>
Dynamic Absorption	para. 3.8	AATCC 70
Soil Release	para. 3.9	AATCC 130, Opt. IV
Toxicity	para. 3.12	<u>4/</u>

- 1/ Visual shade matching. The color and appearance of the finished cloth shall match the standard sample when viewed using AATCC Evaluation Procedure 9, with source simulating artificial daylight D75 illuminant with a color temperature of 7500 (± 200)°K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 (± 200)°K, when tested as specified in 3.5.1.
- 2/ Machine cycle, permanent press; Wash temperature, 120 (± 5)°F; Tumble Dry, permanent press.
- 3/ Wicking Test Procedure. The wicking test procedure is used to determine the rate at which water will wick on test specimens suspended in water.

Apparatus

- Flask. A 500 ml Erlenmeyer flask shall be used.
- Straight Pins. Straight pins approximately 3 inches in length shall be used.
- Food Coloring. A food color shall be used that will be visible on the specimen to indicate the water level during testing.

Testing Procedure A sample size of 6 inches by 1 inch will be cut with the 6 inches cut measured in the Wale direction. A 500 ml Erlenmeyer flask shall be filled with 200 ml of colored water (food coloring used shall be a contrasting color to the cloth material to make the water level visible on the sample) NOTE: Fill as many flasks as specimens to be tested. The top edge of the sample will be pierced with a long straight pin (approximately 1/8 inch to 1/4 inch from top) and the sample then suspended from the top of the flask as shown in Figure 1. After 15 minutes, remove the sample from the flask and measure water level on sample in inches and record. Values are reported as inches per minute.

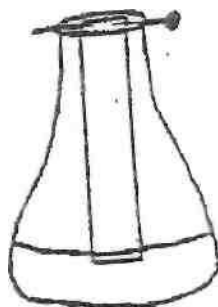


FIGURE 1

4/ A Certificate of Compliance (COC) may be submitted for this requirement.

3.5 Color. The color of the washcloth shall be Brown 436 and shall match the standard sample (see 6.2).

3.5.1 Visual shade matching. The color and appearance of the finished washcloth shall match the standard sample when tested as specified in 3.4.1.

3.5.2 Colorfastness. The finished washcloth shall conform to the colorfastness requirements specified in Table II, when tested as specified in 3.4.1.

TABLE II. Colorfastness requirements

Color Evaluation	Laundering (3 cycles) (min.) <u>1/</u>	Light (40 hours or 170 kJ (min.) <u>2/</u>	Perspiration (acid and alkaline) (min.) <u>1/</u>	Crocking (min.) <u>3/</u>
Brown 436	3-4	3-4	3-4	3.5

1/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change and AATCC Evaluation Procedure 2, Gray Scale for Staining.

2/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change.

3/ Rated using the AATCC Evaluation Procedure 8, AATCC 9-Step Chromatic Transference Scale.

3.6 Dimensional stability. The finished washcloth shall not shrink or elongate more than 4.0 percent in either the warp or filling direction after 5 complete cycles when tested as specified in 3.4.1.

3.7 Wicking. The wicking rate of the finished washcloth shall meet a requirement of 6 inches in a 20 minute time period when tested as specified in 3.4.1.

3.8 Dynamic absorption. The finished washcloth shall show a rating of 75 for initial and 73 after 5 launderings when tested as specified in 3.4.1.

3.9 Soil release finish. The soil release finish of the washcloth shall show a rating of not less than “4” using the AATCC Stain Release Replica, when tested as specified in 3.4.1.

3.10 Labels. The label shall be a white, nonwoven, spun bonded polyester cloth or nonwoven mildew-resistant printed, acrylic coated, impregnated, 80/20 polyester/cellulose cloth, located on one of the outside edges of the washcloth, sewn into the binding on the edge. The label shall be 1-1/2 inches long by 2 inches wide. The label shall contain the item name, contractor’s name, contract number, fiber content, NSN and the following laundering instructions. The information may be printed on both sides of the label.

Laundering instructions

1. Wash with moderately hot water (115°F) using soap or mild detergent.
 - a. For machine washing. Use permanent press cycle.
 - b. For hand washing. Do not wring or twist.
2. DO NOT STARCH, BLEACH OR USE FABRIC SOFTENER.
3. Dry in tumble dryer on warm or delicate setting. Remove immediately at end of drying.
DO NOT REMOVE THIS LABEL

3.10.1 Barcode label. Each washcloth shall be individually barcoded with a paper tag for personal clothing items. The paper tag shall be standard bleached sulfate having a basis weight of 100 pounds. The paper used for the tags shall have a smooth finish to accept thermal transfer and direct printing. The tags shall have a hole and shall be attached to each washcloth by a fastener. As an alternate the barcode label may be a self stick label which shall be attached to each washcloth. The tags shall be clearly legible and readable by scanner. The bar coding element shall be a 13 digit National Stock Number (NSN). There shall be a 12 digit Universal Product Code (UPC) number assigned for all NSNs by the contracting activity. The initial "UPC" must appear beneath the code. The bar code for NSN and UPC type shall be a medium to high density and shall be located so that it is completely visible on the washcloth when it is folded and/or packaged as specified and so it causes no damage to the washcloth. The UPC code shall be placed on all shipping cartons on which the NSN appears.

3.11 Workmanship. The finished washcloth shall conform to the quality of product established by this specification. The washcloth shall be free from all defects which would affect its serviceability, appearance and uniformity.

3.12 Toxicity. The finished washcloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

- a. First article inspection (see 4.2).
- b. Conformance inspection (see 4.3).

4.2 First article inspection. The first article, submitted in accordance with 3.1, shall be inspected for compliance with design, workmanship and dimensional requirements. The presence of excessive defects, as defined by contract (see 6.2) or failure to pass any test shall be cause for rejection of the first article. The sample size shall be specified in the procurement document (see 6.2). The sample unit shall be one (1) washcloth and the lot size shall be expressed in washcloths.

4.3 Conformance inspection. Conformance inspection shall include shade and appearance of all components, finished measurements and defects specified in Table III. Sampling for inspection shall be performed in accordance with ANSI/ASQ Z1.4, as defined in the contract (see 6.2), except where otherwise indicated. The sample unit shall be one (1) washcloth and the lot size shall be expressed in units of washcloths.

4.4 Inspection conditions. In accordance with 4.1, components and end items shall be tested in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable procurement documents (see 6.2). The Government reserves the right to inspect all components and end items to determine conformance to requirements.

4.4.1 End item visual examination. The washcloth shall be examined for the defects listed in Table III. The lot size shall be expressed in units of washcloths. All cloth and washcloth defects shall be scored in accordance with Table III, which are clearly noticeable at normal viewing and affect serviceability and appearance of the washcloth. Material defects are defined in Section 1 of FED-STD-4. If needed, a closer inspection will be performed to verify compliance to purchase description requirements. Shade shall be evaluated at a distance of 3 feet.

TABLE III. End item visual defects.

Examine	Defect	Classification	
		Major	Minor
Material defects	Hole, cut, tear, mend, drop stitch, loose knit, miss knit, slub, streak, knitted in waste, permanent fold, pleat or crease; any spot or stain, discoloration or color not as specified, any shaded part, any objectionable odor	101	
	Shade bar or abrasion mark	102	
	Color not as specified	103	
	Knots greater than Sears Scale Level C (see 2.3)		201
Workmanship	Not specified size	104	
	Design not as specified	105	
	Finished measurements not as specified	106	
Labels	Label(s) missing, incorrect, or illegible; bar code or UPC code label omitted or not readable by scanner; human-readable interpretation (HRI) omitted or illegible; bar code not visible on folded, packaged item; bar code label causes damage to the item.	107	
Cleanness	Grease, oil, dirt or ink stains clearly noticeable	108	

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of the washcloths is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department of Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Departments or Agencies automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The washcloths are intended for use by military personnel of the Department of the Army.

6.2 Acquisition requirements. Acquisition documents should specify the following:

- a. Title, number and date of this Purchase Description.
- b. The specific issue of individual documents referenced (see 2.2).
- c. When first article is required (see 3.1 and 4.2).
- d. Toxicity requirements (see 3.12).
- e. Number of first article samples (see 4.2).
- f. Conformance inspection and acceptance quality limits (see 4.3).
- g. Inspection conditions (see 4.4).
- h. Packaging requirements (see 5.1).

6.3 First article. When a first article is required (see 3.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a pre-production sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisitions documents regarding arrangements for selection, inspection and approval of the first article.

6.4 Standard sample. For access to standard shade samples, address the contracting activity issuing the invitation for bids or request for proposal. Standard shade samples are also available at DLA Troop Support at <http://dscp.dla.mil/Portal/sreqfrm.aspx> .

6.5 A suggested source for the fabric is:

Chima Inc.
1075 Bern Road
Wyomissing, PA 19610-2009
610-208-0904

6.6 Subject term (key word) listing.
Brown 436

Custodians:
Army – GL

Preparing Activity
GL-Army